

Name: AMRITANSU SINGH AMRITANSU SINGH | Reg. No: RA2511003011028

Unit “ 2 : Session “ 8 : SLO “ 2 | Circular Singly Linked List “ Insertion

■ I. Insert a New Node VAL at the Beginning

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 11
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET PTR = START
- Step 6: Repeat Step 7 while PTR -> NEXT != START
- Step 7: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 8: SET NEW_NODE -> NEXT = START
- Step 9: SET PTR -> NEXT = NEW_NODE
- Step 10: SET START = NEW_NODE
- Step 11: EXIT

■ II. Insert a New Node at the End

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 10
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET NEW_NODE -> NEXT = START
- Step 6: SET PTR = START
- Step 7: Repeat Step 8 while PTR -> NEXT != START
- Step 8: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 9: SET PTR -> NEXT = NEW_NODE
- Step 10: EXIT